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ORIGINAL PAPER

Forecasting price parity for stand-alone hybrid solar microgrids: an international comparison

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Abstract This work quantitatively analyzes the evolution of hybrid stand-alone microgrids in the US, Germany, Pakistan and South Africa to determine the grid price parity date for a solar PV/Diesel/Battery hybrid system in each market. The energy system model and NREL's HOMER software are used to simulate the microgrid operation and forecast the levelized cost of electricity into the future for each location, with results ranging from 35–52 cents/kWh. This cost/kWh is compared with projected grid electricity prices at each location to determine parity dates. Analysis results reveal that grid parity is primarily a result of higher power prices or unreliable grids rather than declining microgrid costs. For Pakistan, grid parity is already here for microgrids, while Germany hits parity in the 2023–2040 timeframe. Results for South Africa and the US suggest that places with low grid prices hit parity after the year 2040. Sensitivity analysis results reveal the significant impact of financing on the year of grid parity, particularly in developing markets. We also show that policy measures to eliminate uncertainty and improve financing could push this grid parity years or decades earlier in developing economies where such microgrids may find a greater utility at present or in the near future.

Keywords Grid parity · Microgrid · Stand-alone · Economics · LCOE · System modeling

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1 Introduction

Microgrids are modern, small-scale electricity grids which frequently use renewable technologies like solar or wind to generate power close to the end users and can be connected to the main grid or work as "off-grid" stand-alone systems.

Stand-alone systems often make use of a local renewable energy source in conjunction with storage and a conventional power source (example: diesel generator) to provide reliable power. In recent times, these stand-alone hybrid systems have been used to provide electricity to small communities in rural and remote areas of less-developed countries. With growing applications in the developing world and potential to compete with conventional grids around the globe, these microgrids are evolving into important building blocks of future energy infrastructure planning. However, similar to the case of renewable energy in general, their adoption depends heavily on economics. As the technology evolves, one may expect the associated energy costs to fall gradually, similar to emerging technologies like solar photovoltaics (PV) or wind turbines [1]. At the same time, grid electricity prices in most regions of the world are expected to increase [2]. These two effects lead to an important question: When will off-grid microgrids reach price parity with grid electricity?

Microgrid price parity is most commonly defined as the milestone when the cost of power from a microgrid falls to the price of conventional grid-supplied electricity [3]. This parity is important because many argue that once consumers are offered environmentally friendly, reliable electricity produced at an equal or lower price than that from the grid, they would overwhelmingly choose the cleaner, more reliable option.

In existing literature, the few cost parity studies for renewable-based microgrid systems are limited to a single country and mostly focus on a grid-tied system. Parity research for multiple countries reported by [4,5] not only use assumptions which fail to internalize diversity across different markets, they also rely on mathematical models which do not capture the time series operation of an off-grid system. Even though [6] conduct a techno-economic analysis to model an off-grid system and determine grid parity for various markets within the United States, their definition of parity, use of optimistic interest rates and other assumptions skew the final results, concluding that microgrids are currently or will soon be cheaper than grid electricity in most of the US.

In general, such parity studies show a divided opinion over concepts like "cascading grid defection" and the "utility death spiral" which state that the economics of off-grid systems will soon become favorable enough for customers to self-supply electricity, leave the grid, cease paying their portion of the fixed costs, and push up prices for other utility customers, causing further "defection". Studies like [6] conclude that solar + battery grid parity is already here or coming soon for utility customers in the US, which could trigger a revenue decay for utilities, eventually leading to the demise of traditional utility business models. Others, like [7], caution that grid defection is expensive relative to the low costs of grid electricity in many developed regions.

This work contributes to this evolving discussion about the true costs and benefits of off-grid systems in competition with grid electricity, providing a comparison of the economics of grid defection in four different locations. We conduct a techno-economic analysis of a stand-alone solar PV/diesel generator/lead-acid battery hybrid microgrid and estimate grid parity date in four energy markets: Central United States (Columbus, Ohio), Germany (Munich), Pakistan (Hyderabad), and South Africa (Johannesburg). NREL's HOMER [8] and the energy system model (ESM) [9] are used to model the operation of such a hybrid setup in each of the four considered energy markets during the 2015–2040 period. The use of two different modeling tools helps compare and validate parity estimates for each location as well as account for modeling uncertainty. Using our results, we compare microgrid competitiveness across the four studied locations and discuss the implications of this parity for policy.

2 Material and methods

For this research, an off-grid PV/diesel/battery hybrid microgrid system is considered. A hybrid energy system generally consists of a primary renewable source working in parallel with a standby secondary non-renewable generator and energy storage units [10]. A hybrid system, as suggested by [10], offers a potential solution to the problems of solar/battery systems like low capacity factors, excess battery costs, and limited capacity to store extra energy [11]. Wind power is not considered as an alternative to solar in the hybrid system in order to limit the scope of the already-complex analysis and provide greater depth on solar, which is a more likely technology at the smaller scales considered for the analysis.

The energy system model (ESM) and Hybrid Optimization Model for Electric Renewables (HOMER) modeling software are used to simulate the operation of a microgrid to determine a system design that minimizes Levelized Cost of Electricity (LCOE) for each location at 5 year intervals into the future. These minimized energy cost estimates result from an iterative evaluation of distinct combinations of system configuration using solar PV, battery and the diesel generator such that the resultant system design has the lowest LCOE from all possible combinations of solar PV/battery/diesel (including solar only, solar + battery only, solar + diesel only). A residential system in the US, for example, is more likely to be only solar + battery. However, such a system would not cost less than the hybrid systems examined here.

The microgrid results are affected by prices of fuel, solar PV, batteries and installation/labor, which vary across the four countries due to differing policies and markets. Distinct forecasts of future costs and operational parameters for each location help determine LCOE values into the future. Grid parity is then defined as the point of intersection between the LCOE trend lines and the edges of a 'parity region' between the optimistic and pessimistic estimates of retail power prices for each location. This 'parity region' acknowledges uncertainty in future price estimates (and parity) and allows to predict a range of years when parity can be expected at a particular location. Subsequently, throughout the study we express results in the form of a parity date range rather than a central figure. The approach is contrary to the work of [6] where, despite using a similar concept, parity is defined as the discrete point of intersection between the highest grid price estimate and the LCOE trend line, skewing the basis for any discussion of results towards the earliest possible date. Moreover, unlike the work of [6], the use of two different microgrid modeling tools with different search algorithms helps validate results since the study involves a prognosis of many parameters, most of which are based on uncertain assumptions.

Table 1 The four countries and respective locations considered in the analysis

Country	Location
United States	Columbus, Ohio
Germany	Munich
Pakistan	Hyderabad, Sindh
South Africa	Johannesburg

2.1 Levelized cost of electricity (LCOE)

The LCOE is a primary metric for the cost of electricity produced by a generator over its lifetime. It is determined by dividing the discounted total costs by the time-discounted energy generated. These include all lifecycle costs including initial investment or capital costs, operations costs and cost of fuels. It is typically used to compare relative costs of energy produced by different sources. Units are typically cents/kWh or $/kWh. The following mathematical relation defines the LCOE:

$$LCOE = \frac{\sum_{t=1}^{n} \frac{CC_t + M_t + F_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{E_t}{(1+i)^t}}, \tag{1}$$

where CC_t are the capital costs in the year t; M_t are the operation and maintenance costs in the year$t$; F_t are the fuel expenditures in the year t; E_t is the energy generated in the year$t$; n is the life of the system and i is the interest rate.

2.2 The four locations

The four countries considered in this comparative study are the United States, Germany, Pakistan and South Africa. They are chosen to maintain diversity in both economics and geography, while ensuring the availability of relevant input data. Of the four, the United States and Germany are both developed countries with mature energy markets, ample energy resources, and good policy frameworks. However, they are significantly different in grid electricity costs and their use of renewable energy, especially solar PV. In contrast, South Africa and Pakistan are developing economies with strained energy sectors and weaker electricity system governance. Their energy sector problems, along with good renewable energy resources make them interesting potential candidates for microgrid solutions. Details on the energy sector situation in each market are presented in Appendix A.1 (pp. 1–3). One specific location in each country, shown in Table 1, is considered for the analysis.

2.3 The energy system model (ESM)

The ESM is an engineering-economic model that inputs a particular system configuration, electricity load time series and solar resource time series to determine the time-varying operation of each component. It calculates the LCOE with other relevant financial information for each considered system [9]. Within input constraints, the model iterates to choose the LCOE-minimizing system design under the given set of parameters (such as PV costs or diesel prices) and can be used to study how changes in these parameters affect the optimal system configuration. It is used to model the operation of the stand-alone system under consideration to determine the LCOE for the optimal system given the inputs at each location and time period. The output of the model includes time series data for each of the system components (i.e. PV, diesel generator, batteries) and the corresponding financial information for the chosen optimal configuration. These include net present costs for fuel, generator, PV and battery as well as LCOE. Further details on the ESM are presented in Appendix A.2 (pp. 3–4).

2.4 NREL's hybrid optimization model for electric renewables (HOMER)

Hybrid optimization model for electric renewables or HOMER, created by the National Renewable Energy Laboratory (NREL) in the US, is also used in this study to validate results obtained from the ESM. HOMER is a general-purpose hybrid system design software that facilitates the design of electric power systems for stand-alone applications [12]. It simulates a PV/Diesel/Battery hybrid system based on the hourly electric load data profile and the solar irradiation data of a specific location over a period of 1 year. Therefore, similar to the ESM, HOMER takes as inputs three forms of data: estimated electric load data in the form of load profile time series, environmental/climate data such as the annual solar resource profile, and financial/technical data for system components.

Based on these inputs, it performs hourly simulations to determine how different systems can be used to meet the load. With the input information and the choice of component sizing and pricing, HOMER is able to simulate the most economical and technically feasible solution at a specific location [13].

All of the scenarios examined in this work were investigated with both ESM and HOMER. Because the results were generally qualitatively similar, discussion mostly focuses on the ESM results. Since both tools have very similar input formats, data conditioned for either one can be used for the other with minimal reformatting.

3 Calculation

3.1 Modeling the stand-alone hybrid system

Since both the models have a fairly large number of input parameters, data from multiple sources is gathered and prepared to serve as inputs for all locations to limit bias and reduce uncertainty. A combination of various economic and technical inputs helps define the constraints on a microgrid system and affect calculations for the LCOE. Figure 1 summarizes the inputs and outputs obtained from the model.

Other technical inputs for both models are set at default values and kept constant throughout the analysis. These have been tabulated in Appendix A.3 (p. 5). In each location, the model is run at 5 year intervals with input values for the years 2015 till 2040. Economic inputs are updated for each of the six optimization runs at a particular

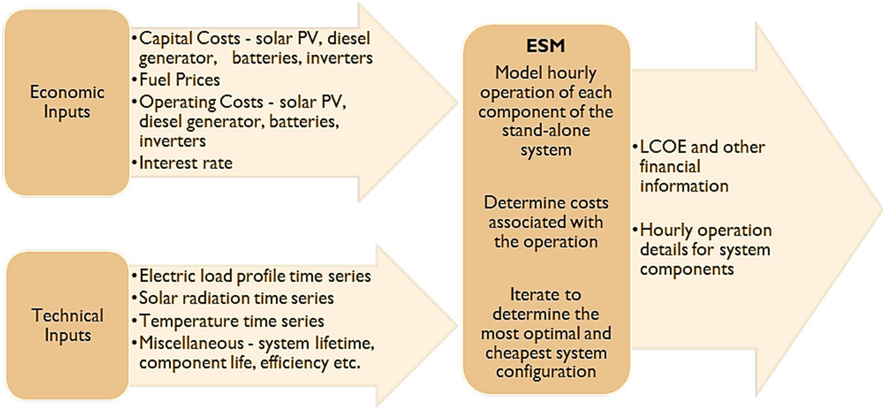

Fig. 1 Relevant economic and technical data inputs define the constraints on the stand-alone microgrid to be modeled in the ESM. Most inputs are location specific, differentiating the four markets. The ESM models the hourly operation of the system and outputs the LCOE and other details. HOMER operates in a predominantly similar way

location, while some of the technical system parameters are kept constant for the entire analysis at that location. Between locations, both technical and economic inputs are updated. The 5-year interval is applied five times at each location to render LCOE values of optimal system configurations for the next 25 years. These results can be compared with projected grid electricity prices, which are gathered and estimated through a separate process described below.

3.2 Data inputs to modeling

Technical inputs, like component lifetimes and generator fuel consumption curves, are kept constant for different locations in order to maintain system-level consistency across the four locations, allowing direct comparison of the results between locations. These inputs are tabulated in Appendix A.3 (p. 5). Other basic parameters, presented in Table 2, are varied along two dimensions: time and location.

All values are in real US 2014 dollars. Finances are from the consumer's perspective rather than a social perspective, and costs reflect prices to end consumers (with subsidies) unless mentioned otherwise. This approach is adopted because the goal is to determine when microgrids will be a preferable economic choice for consumers. A similar approach is adopted by [6] in their study of the US market. Moreover, the economic analysis assumes a consumer with existing access to grid power. For unserved consumers in developing countries like Pakistan and South Africa, the costs of grid-extension would have to be accounted for in grid price calculations. This is beyond the scope of this study and forms the basis for a different research question on the evaluation of grid-extension versus stand-alone grid options.

Current and future cost estimates for system components such as the generator, solar PV, batteries, inverters and diesel fuel prices make up essential inputs to the models. The following section provides an overview and summarizes the rigorous

Table 2 The list identifies ESM input parameters varied in space and/or time for the analysis

Input parameter	Varied in time	Varied across locations
Generator capital costs	No	Yes
Diesel fuel prices	Yes	Yes
Solar PV capital costs	Yes	Yes
Battery capital costs	No	Yes
Inverter capital costs	Yes	Yes
Generator operational costs	No	No
PV operational costs	No	Yes
Installation costs	No	Yes
Discount rate	No	Yes

analyses undertaken to carefully arrive at cost estimates for these model inputs. While many current values are taken from credible (and usually multiple) sources to reduce bias, future forecasts are determined using a variety of methods ranging from simpler techniques like data extrapolation and averaging, to more complex approaches like logarithmic learning curves. For brevity, most details and mathematical working have been presented in the appendices, which have been extensively referenced within the text.

3.2.1 Input parameters

Throughout the analysis, real 2014 US dollars are used and constant exchange rates assumed for currency conversions as presented in Appendix A.4 (p. 5). For cases where data is available in nominal dollars, a relevant inflation rate is used for conversions (Table 3). The discount rate is another important parameter governing present value calculations and a determinant of the final optimal system configuration. This is because the technologies being used are different in terms of their distribution of expenses: while solar PV has high capital costs, the largest share of costs for a diesel generator are spread in time as fuel expenses. Since discount rates are difficult to set, typically the cost of capital is used as a proxy for the discount rate [14]. Therefore, cost-of-capital figures for different markets are relatively easily available and are frequently used in present worth analyses. We use cost of capital rates in this analysis whenever discount rate estimates are unavailable.

For solar PV costs, a separate a learning curve analysis is used which allows uniformity in solar price projections. Learning curves describe how costs decline with cumulative production, where the cumulative production is used as an approximation for the accumulated experience in producing and employing technology [30]. This decline in cost is a constant percentage with each doubling of the total number of units produced, characterized by the learning rate LR. The learning curve equation is written as:

$$C_t = C_0 \cdot \left(\frac{P_t}{P_0}\right)^b, \tag{2}$$

Table 3 Summary of base case input parameters with short descriptions of the sources/methods employed

Input parameter	US	Germany	Pakistan	South Africa
Inflation rate (%)	2%	2%	10.9%	6.5%
	For the US, the 2% figure is an average for the last decade [15]. The 2% inflation rate used by both [5,16] is used in Germany. The average inflation rate for the 2009–2013 period reported by [17] is used for South Africa. For Pakistan, the 2013 Pakistan Economic Survey (Government of Pakistan, 2012) inflation figure is assumed			
Discount rate (%)	8%	5%	14%	10%
	Figure for the US is based on assumptions from [19,20]. For Germany, it is based on the 4–5% range used by [16], a 5% figure used by [21], a 3.6% cost of capital rate estimate for commercial scale solar PV determined by [5] and a 7% figure used for all countries across Europe by [22]. Ondraczek et al. [23] estimate the cost of capital rate for Pakistan at 13.8%, which is used. In the case of South Africa, [24] use a 10% rate, while South Africa's power regulator (NERSA) and largest utility (Eskom) both assume an 8% real rate for utility scale power [25]			
Current solar PV capital cost (2014\$/W)	3.80	2.15	2.45	3.35
	These are used as a starting point for the learning curve analysis. For the US, the recent SEIA/GTM research report, [26] helps determine current system cost estimates while for Germany, [27] provides this figure. In their report, [28] provide a good estimate for system prices in South Africa. For Pakistan, solar PV system prices are obtained from a solar systems entrepreneur company, T.S.K Engineering International (Pvt.) Ltd			
Solar PV learning rates (%)	20%	10%	15%	15%
	For the US, considering greater solar PV penetration, favorable policies and ample potential for market consolidation [29], a learning rate higher than 14% is applied. With a strong potential for growth given good solar resources and troubled energy markets countered by a lack of supportive polices, the use of an average learning parameter for both the developing countries is plausible. For a mature and consolidated PV market in Germany [29], the highest progress ratio is assumed. This method is previously mentioned by [30] , where they make note of a similar approach for individual countries			
Battery costs (2014\$/Wh)	0.160	0.219	0.170	0.216
	Lead-acid battery costs are determined by averaging battery prices from multiple suppliers in each market and ignoring any installation costs (which are accounted for separately)			
Inverter cost share x (%)	8%	11%	5.6%	13%
	For the US, figure reported by [6] is used. For Germany, it is estimated using system cost figures from [29] . For South Africa, the system cost breakup provided in [28] is used. For Pakistan, T.S.K Engineering's quotation is used. This percent figure is assumed to stay constant for each location over the entire period of the analysis			
Diesel fuel (2014\$/L)	For the US, current and future diesel price figures are obtained from [26]. For oil importing Pakistan and South Africa, where oil prices are linked to the international market, future US price trends are assumed. Following from the work of [16], future fuel prices for stationary power applications in Germany are based on the heating oil price forecast provided by [31]. In Germany, "heating oil" has lower taxes and is bought in bulk amounts for stationary applications [32]. Details presented in Appendix B.1 (pp. 5–6)			

Table 3 continued

Input parameter	US Germany Pakistan South Africa
Installation costs (%)	An additional 5% cost is applied to the total system capital cost at all locations to account for civil works, wiring, and any overlooked factors (e.g. battery installation costs)

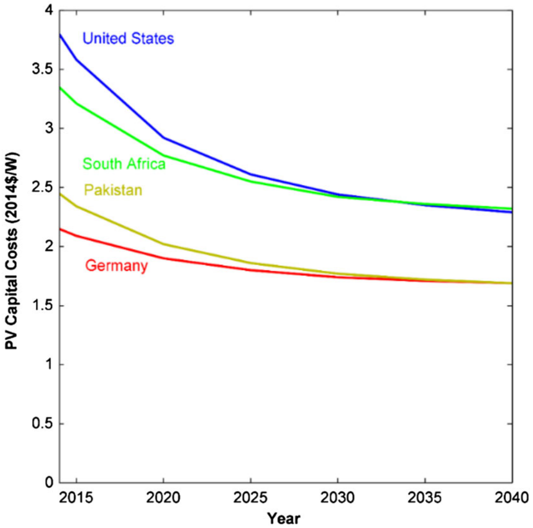

Fig. 2 Future projection of installed solar PV system costs based on the learning curve analysis and used in this work. The costs include PV modules, BOS components (including the inverter) and installation costs

where C_t is the cost of cumulative production at time t; C_0 is the cost at initial level of production at $t = 0$; P_t is the cumulative production at time t; P_0 is the cumulative production at $t = 0$ and b is the learning parameter defined as $\frac{\log(1-LR)}{\log 2}$.

The results of the learning curve analysis are shown in Fig. 2. Current system price in the US is significantly higher, which can be attributed to differences in non-hardware costs, as investigated by [29]. However, with improved system learning, market consolidation and standardization the US price is expected to approach but not reach that of Germany (Fig. 2). Detailed mathematical working for the complete analysis is presented in Appendix B.2 (pp. 6–12).

Solar PV replacement costs are kept equal to the initial capital cost figures in each case. The operational costs for solar PV are kept at 1% of the system capital costs for the year 2015. Both [13,30] use the same 1% figure in their calculations. In their report, [33] find that the yearly maintenance costs for solar PV lie in the 0.5–1% range. However, for subsequent years, this operational cost figure is kept constant in real 2014 dollars rather than decline with installed cost. A fixed value rather than a fixed percentage is used to neglect any learning effects that may push down the maintenance costs to unrealistically low levels.

Diesel fuel prices for each location are carefully conditioned to omit transportation fuel taxes, which are normally removed or refunded for non-transportation uses. Generator replacement costs are kept equal to the initial capital costs in each location.

Both the upfront and replacement costs are assumed constant in real 2014 dollars over the entire period of the study. Since the capital costs for a generator are very small compared to the operating costs of fuel, this assumption has little effect on the final optimization results. Generator cost determinations are presented in Appendix B.3 (pp. 12–14).

As the incumbent energy storage technology, lead-acid batteries are used in this analysis. It is typically (often implicitly) assumed that learning in lead-acid battery production is “finished”. The literature analyzing the price-point goal for emerging energy storage technologies refers to a static value of current lead-acid battery prices [34]. Based on this and the limited amount of data available on storage device price projections, as a conservative estimate, current battery prices are assumed to stay constant in real 2014 dollars throughout the analysis. Actual working is presented in Appendix C (p. 14).

Inverter costs are derived from the PV system costs using Eq. (3):

$$Cost_{inv} = x \times Cost_{PV}, \tag{3}$$

where $Cost_{inv}$ is the invertor cost input; x is the inverter costs expressed as a percentage of the solar PV system costs (Table 3) and $Cost_{PV}$ is the solar PV system costs determined from the learning curve analysis. The same approach is used by [6] and is a plausible reflection of relevant available data. Inverter replacement costs are considered equal to the initial capital costs for each location.

3.2.2 Time series data

Both the ESM and HOMER take as inputs, hourly time series data for the solar resource and the electric consumer load profile at a given location for one complete year. Even though the ESM is capable of modeling at sub-hourly resolutions, time series data used in this study is for a complete year at 1-h resolution, resulting in 8760 data points in each case. The 1-h resolution provides a single basis for comparison of results between HOMER and ESM, because HOMER uses a time resolution of 1 h for system operation [9].

Solar radiation data for each location is extracted from HOMER’s built-in solar resource data function. Based on the longitude and latitude of a location, HOMER accesses NREL’s online databases that serve data from either NREL’s Climatological Solar Radiation (CSR) or NASA’s Surface meteorology and Solar Energy (SSE) data set [35]. The resulting data is an hourly time series for an entire year and can be exported for use elsewhere.

Hourly electric load profile data defines the annual electricity consumption at a 1-h resolution and is essentially the target load that must be met by the hybrid microgrid. For this study, electric load data characteristic of each location is used. Except for the US, residential load patterns for the other three locations are processed from raw data because 1-h resolution data is not readily available. For the US, 1-h resolution data is available and used as-is. Details of residential electricity load data conditioning are presented in Appendix D.1 (pp. 14–15).

3.3 Utility electricity price projections

Grid power prices at each location are used to compare and determine grid parity, where grid parity is defined as the point when grid power prices become equal to the LCOE for the hybrid stand-alone system. Two distinct grid price forecasts are used to define a 'parity region' into the future. This grid parity area, enclosed by the upper and lower grid power price forecasts, helps provide a possible range of grid power prices and hence grid parity dates at each location. Since retail electricity prices are a function of many socio-economic, technical and policy parameters and vary greatly between different markets, the use of a 'parity region' is a more realistic reflection of uncertainty in future price estimates and consequently, parity. Providing a range rather than point estimates is a more appropriate result for this study given the uncertainty and assumptions involved in the optimization. The 'high' and 'low' retail prices help define optimistic and pessimistic parity dates at each location. The two grid price projections defining the parity region are based on the extrapolated recent trend of prices from the last 8–10 years in each market and the future grid power prices estimated by the respective governments or regulating bodies. In all cases, extrapolating the recent trend implies more rapidly increasing prices than the government estimates. All price figures are in 2014 dollars, so that the parity region reflects the change in real prices over time. While mass grid defection would affect grid electricity prices, this is not considered in our work because (a) its effects are complex and difficult to predict, and (b) we are seeking the point when consumers are likely to begin switching over.

For the United States, historical trends (2006 onwards) and future forecasts of grid power prices from the EIA AEO 2014 are used. For Germany, a recent trend of residential price data available on [36] is used to extrapolate a future price trend line. Government estimates are obtained from [31]. For both Pakistan and South Africa, multiple studies are used to determine future trends. Comprehensive details of these calculations for grid parity regions are presented in Appendix E (pp. 16–29).

Pakistan and South Africa currently face frequent power outages due to shortage, while the modeled microgrid alternatives have significant excess generation capacity to ensure reliability. In order to account for and monetize costs associated with grid unreliability, we estimate the additional costs of achieving reliable power while connected to the grid. This cost of reliability is incorporated within the utility price forecasts for both Pakistan and South Africa. We assume that the electricity grids in the US and Germany are sufficiently reliable for direct comparison to the microgrid systems.

In an analysis separate from the microgrid estimation, ESM is used to simulate the unreliable grid and determine the cost to produce substitute power during outages. This estimate is used as a monetary measure of the cost of reliable power in the analysis. A similar approach is used by [37] to model an unreliable grid for grid-connected distributed generation. They model the unreliable grid in HOMER, using a diesel generator module with scheduled downtime. For this study, both grid-generator and grid-battery setups are separately modeled in the ESM to determine the cost of substitute power from a generator and batteries during outages. A comparison of this cost of substitute power from a generator and batteries at each location for the year 2015 shows the generator to be the cheaper option. Therefore, a grid-generator back-up is considered for the base case analysis.

An intermittent solar resource based on the load shedding schedule is used to emulate the unreliable grid so that the generator substitutes to fill in whenever solar PV (grid) is out. With only cost figures for the generator input to the model, the resulting LCOE value is then defined as:

$$LCOE_{ESM} = \frac{Cost_{gen}}{kWh_{gen} + kWh_{grid}}, \tag{4}$$

where $LCOE_{ESM}$ is the LCOE result of the simulation (ESM output); $Cost_{gen}$ is the cost of substitute energy supplied by the generator during an outage; kWh_{gen} is amount of substitute energy supplied by the generator (ESM output); and kWh_{grid} is the energy available from the grid (ESM output).

Using the above relation and ESM outputs, the $Cost_{gen}$ is determined as follows:

$$Cost_{gen} = LCOE_{ESM} \times \left(kWh_{gen} + kWh_{grid}\right). \tag{5}$$

The cost of available grid power, on the other hand is determined using the following relation:

$$Cost_{grid} = Price_{grid} \times kWh_{grid}, \tag{6}$$

where $Cost_{grid}$ is the total cost of energy available from the grid and $Price_{grid}$ is the price of unreliable grid power in $/kWh.

Using results from Eqs. (5) and (6), the final grid price estimate with the additional cost of substitute power included is then determined by the relation:

$$Cost_{grid/gen} = \frac{Cost_{grid} + Cost_{gen}}{kWh_{gen} + kWh_{grid}}, \tag{7}$$

where $Cost_{grid/gen}$ is the estimated resultant cost of reliable power using a generator back-up.

The detailed mathematics of these calculations along with parity region and battery back-up results are presented in Appendix E (pp. 16–29).

Figure 3 shows the grid parity regions for all four locations. For Pakistan and South Africa, the figure shows LCOE with and without this cost of substitute power from a diesel generator. Because power outages are more frequent in Pakistan than in South Africa, the effective cost of unreliable electricity in Pakistan is much higher than the corresponding price projection without the cost of substitute power.

4 Results

As discussed earlier, grid parity is defined by the intersection of the LCOE trend line with the two edges of the parity region. There is considerable uncertainty in these parity results due to various assumptions and limitations of the forecasts. However, the use of a parity region, two separate modeling tools, and detailed sensitivity analysis help to identify the effects of uncertainty and provide a range of grid parity dates at each location.

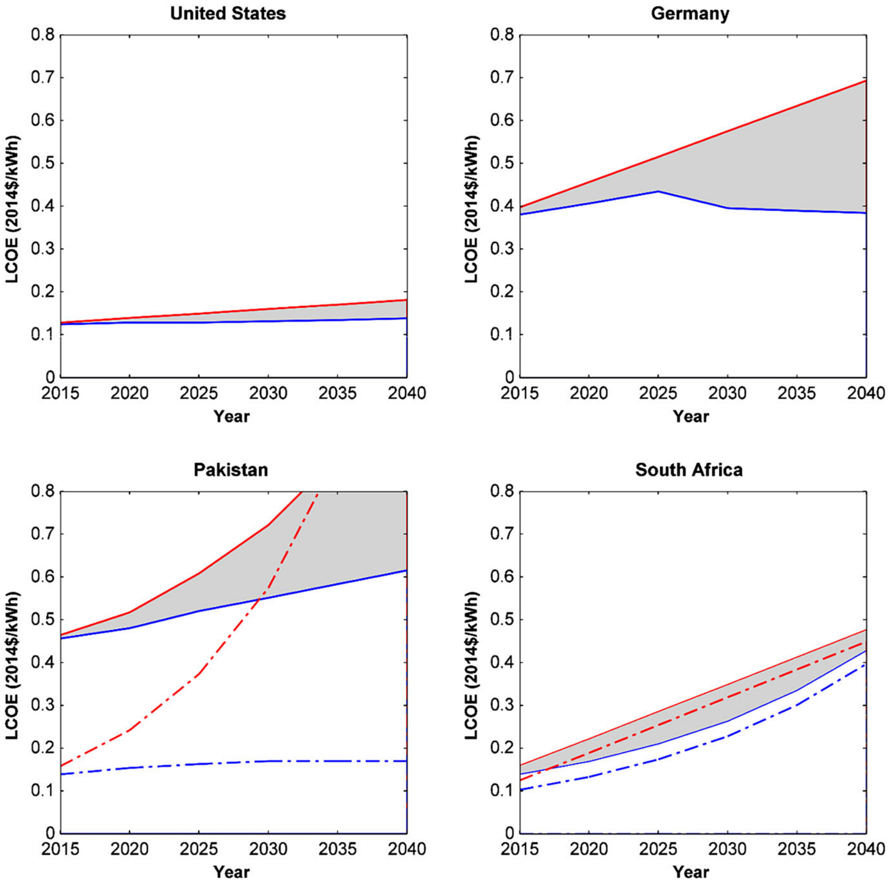

Fig. 3 Projected grid electricity price regions for the four studied locations. Each region is contained by two edges, highlighted in *red* and *blue*. The *red edge* corresponds to future estimates of grid prices obtained from extrapolated recent price trends. The *blue edge* shows government projections of grid prices in the respective market. For Pakistan and South Africa, the *solid regions* also include estimates of the cost of unreliable grid power by including a generator for backup power. The *dotted regions* trace future grid prices without the additional costs of backup power. In both cases, estimates of the effective cost of unreliable grid power are higher than future grid price estimates due to the additional cost of substitute power. The difference in these two estimates is greater for Pakistan due to a higher number of annual power outages compared to South Africa (color figure online)

Figure 4 shows HOMER and ESM LCOE results for the four locations. For detailed LCOE data and the corresponding system configuration details refer to Appendix F (pp. 30–32).

As seen from Fig. 4, the complete set of LCOE values obtained from both the ESM and HOMER lie within the 35–55 cents/kWh range. The increase in LCOE values over the study period is less than 10 cents/kWh. Given the level of uncertainty associated with the optimization, simplifying assumptions used in input data conditioning, and the length of the study period, this increase is small compared to the uncertainty. Corresponding results for each year from both HOMER and the ESM are generally within 10 cents/kWh of each other, with an average difference of 4 cents/kWh.

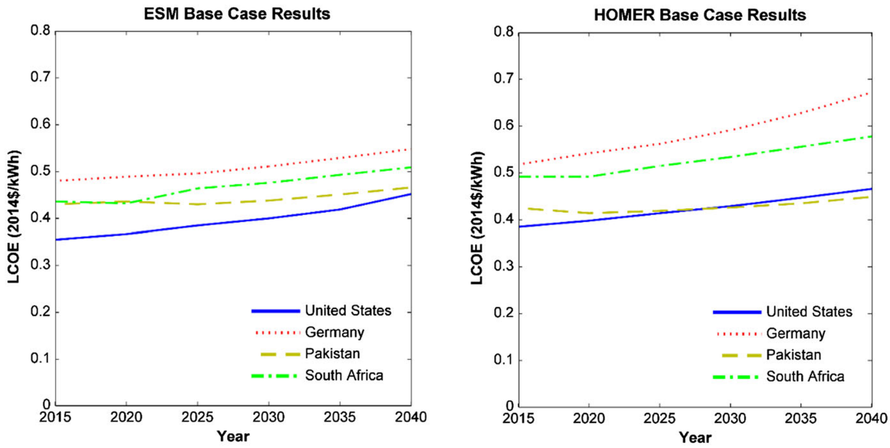

Fig. 4 HOMER and ESM LCOE results for all four locations. HOMER results are between \$0.37/kWh and \$0.66/kWh while the ESM results vary between \$0.35/kWh and \$0.55/kWh. In general, HOMER results for each country are slightly higher than the corresponding ESM LCOE values. In both cases, however, the relative ranking and trends are consistent. Results from both tools follow a general trend of gradual increases in LCOE for each country

These results suggest that lowest-cost microgrid system costs will increase over time, which is unexpected for an emerging technology. Examining the data in detail reveals that LCOE values in each case follow a trend similar to that of diesel prices in the respective country. This is because the optimal system in each case has a large share of diesel generator as the lowest-cost approach to meeting demand. The decline in solar PV prices and increased savings due to this price decrease do not offset the increased cost of diesel. This is true even as the optimal system shifts from diesel generation to greater reliance of solar PV over time. However, due to a significant share of renewables in the optimal systems, the overall increase in LCOE values is smaller than the increase in diesel prices over the 25-year study period. In other words: while a solar PV/battery system would see continual price decreases over time, the lowest-cost option tends to use a significant amount of diesel generation, which gets more expensive over time. This effect can be seen from ESM base case results for the US, illustrated in Table 4.

As expected, when diesel prices rise, both ESM and HOMER determine that the optimal system uses less diesel generation and more solar/battery. Germany has the highest diesel fuel prices (Appendix B.1 p. 6) and therefore the highest LCOE values, while the US has the lowest in both cases. Optimal systems for Pakistan, Germany and South Africa have relatively higher solar PV fractions corresponding to conditions favorable to solar (i.e. good solar resource or low PV prices).

Figure 5 shows LCOE estimates with the respective parity regions in each market. The United States has the lowest LCOE figures for the solar hybrid microgrid system. With the lowest diesel fuel prices and significant government incentives for solar PV, LCOE values are within the 30–40 cents/kWh range. However, a decrease in solar prices and an increase in diesel prices is not enough to offset the significant difference between solar PV and diesel. Therefore, over the entire period of the analysis, the

Table 4 ESM results for the US

United States	2015	2020	2025	2030	2035	2040
ESM LCOE $/kWh	0.355	0.367	0.385	0.400	0.419	0.452
Solar Fraction %	30.6	28.5	31.8	32.7	34.5	40.0
PV (kW)	154.3	134.2	160.9	174.2	181.8	217.6
Generator (kW)	111.3	110.9	111.4	110.8	110.7	128.8
Battery (kWh)	134.4	151.4	168.6	146.1	226.7	541.4
Solar PV Cost ($/day)	105.5	99.2	106.2	107.3	107.5	125.9
Generator Cost ($/day)	17.8	17.8	17.8	17.7	17.7	20.6
Battery Cost ($/day)	16.8	25.3	32.4	20.9	60.2	115.6
Diesel Cost ($/day)	412.5	432.3	444.6	479.4	468.2	436.3

The optimal system is a diesel generator dominated system. The solar fraction increases somewhat with decreasing solar PV prices. However, the decrease in prices is offset by an increase in diesel fuel prices, resulting in net LCOE increases. Similar results for other places are presented in Appendix F (pp. 30–32)

diesel generator remains the dominant system element. HOMER LCOE results follow a similar trend, staying within an average 3 cents/kWh of the ESM results. Owing to low grid electricity prices in the US in general, and Ohio in particular, an off-grid hybrid microgrid does not come close to grid parity during the analysis period.

Germany has relatively high diesel fuel prices due to taxes (other than transportation taxes) levied to push for cleaner fuels. On the other hand, their solar PV prices are one of the lowest in the world due to pro-solar policy initiatives. However, due to the poor and seasonally-varying solar resource, the optimal system is always forced to use a diesel generator to meet load. As a result, the average LCOE for Germany turns out to be 15% higher than the combined average for the other three countries. Even though the system has a high LCOE, it hits grid parity around the year 2023 if grid electricity prices follow the recent trend. Since 2011, government taxes, levies and fees have amounted to about 50% of the nominal price per kWh paid by residential customers in Germany [38]. The trend means an aggressive price hike, pushing the parity point closer in time. For HOMER, this parity estimate is around the year 2034 due to higher LCOE estimates. Both of these parity points occur at the 'Recent trend' edge of the parity region which implies that, unless grid prices in Germany continue to rapidly increase, a stand-alone solar PV/diesel/battery hybrid microgrid does not hit cost parity with the grid during the study period.

Given the best solar resource amongst the four locations, the optimal system for Pakistan maintains a solar PV fraction comparable to systems in Germany, despite much lower government support for the technology. This suggests a decent balance between solar and diesel, reflected in the final LCOE results for Pakistan where the rise in LCOE is only around 3 cents over the entire study period. LCOE values lie between the German and US LCOE results. The analysis suggests that grid parity is already here for Pakistan (Table 5). With increasing electricity prices and an unreliable grid characterized by 8–10 h of power outages on an average day, microgrids make economic sense in Pakistan at present. The parity region shown in Fig. 5 accounts for the retail price of electricity and the cost of unreliable grid to the end consumers.

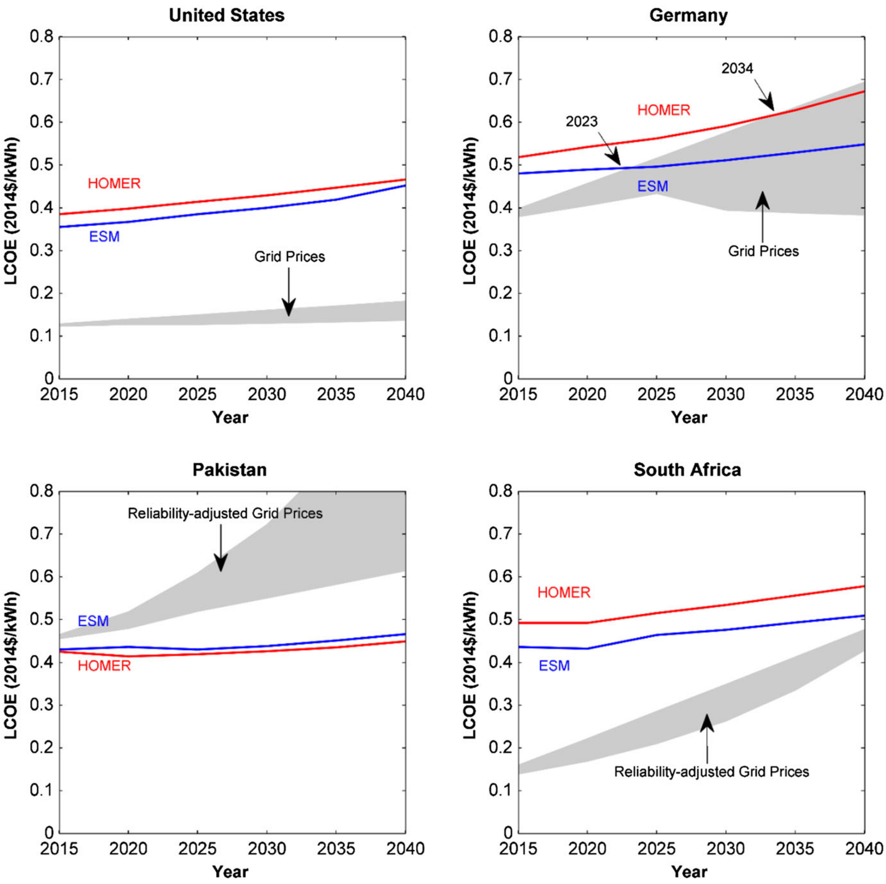

Fig. 5 HOMER and ESM LCOE results with future grid electricity price estimates defined by the shaded parity region for all four locations in real US 2014 dollars. Grid parity point estimates are defined by the points of intersection between the edges of the parity region and the LCOE lines. So, while grid parity does not occur for the US and South Africa during the study period due to low grid electricity prices, grid parity for Germany occurs in the years 2023 and 2034 for ESM and HOMER results respectively. Both these points are on the extrapolated recent trend edge of the parity area, reflecting when the 'earliest' parity date would occur, provided recent increases in real grid electricity prices continue in the future. For Pakistan, grid parity is already here, given the high cost of unreliable grid power

The optimal system configurations in South Africa turn out to be diesel-dominated systems with the solar fraction accounting for just above one third the energy supplied. Therefore, the final LCOE results from both HOMER and the ESM follow a gradual increasing trend, similar to that of the diesel prices. However, due to these rising diesel prices, the solar PV fraction gradually increases in later years. In comparison to Pakistan, with more expensive solar PV and battery in South Africa, this increase in the fraction translates to higher LCOE values. As seen in Fig. 5, microgrids do not hit parity in South Africa during the study period, due partly to currently low grid electricity prices.

Table 5 Base case grid parity point estimates for all four locations

Base case parity estimates		
Location	Earliest	Latest
USA	Past 2040	Past 2040
Germany	2023	Past 2040
Pakistan	Before 2015	Before 2015
South Africa	Past 2040	Past 2040

While grid parity is already here for Pakistan, it does not occur for the US or South Africa during the study period. For Germany, grid parity occurs during the study period in the year 2023 only if recent electricity price hikes continue and past that point if rates rise at a lower rate

Figures 6 and 7 show sensitivity analysis results for the cost of capital rate and diesel fuel prices. In both cases, 'high' and 'low' values for the input parameter are used relative to those from the base case. For the cost of capital rate sensitivity, different rates are used at each location because the base case values at the four locations are very different. The use of relative values helps determine the upper and lower bounds for the LCOE values and thus highlights the sensitivity of grid parity points to these inputs at the particular location.

As expected, higher rates translate into higher LCOE values. However, the resulting changes in LCOE are limited to less than 10 cents/kWh for all locations. Moreover, the cost of capital rate governs the composition of the optimal system, with an increase in the fraction of solar PV and battery for lower rates. As discussed earlier, this is because of the financial difference between the two technologies being used in the hybrid system, where solar PV has greater upfront costs, while diesel generation is costlier to operate. In all cases, a higher cost of capital rate pushes the parity points farther in time whereas lower rates imply microgrids hit parity sooner. However, in most cases this change in parity point is limited to 5 years. System configuration and LCOE details are presented in Appendix G.1 (pp. 33–35).

For diesel price sensitivity, 'high' diesel price estimates are based on a real price increase at a rate twice the current (base case) rate, while the 'low' estimates assume no real increase in prices. For each location, higher fuel prices result in higher LCOE values and vice versa. The difference in LCOE between these alternative scenarios and the corresponding base case results gradually increases to around 20% in the year 2040 for all locations, as seen in Fig. 7. The inverse trend is observed with lower diesel prices. Therefore, Fig. 7 depicts LCOE lines 'fanning' out from the year 2015. The final optimized system configurations for both the 'high' and 'low' diesel price cases are not significantly different from the base case, with a few exceptions. Details are presented in Appendix G.2 (pp. 35–38).

5 Discussion

The LCOE metric helps to capture the considerable variation in geographic and economic conditions across different markets because the input parameters to the LCOE are mainly dependent on circumstances regarding geography, time, energy and finan-

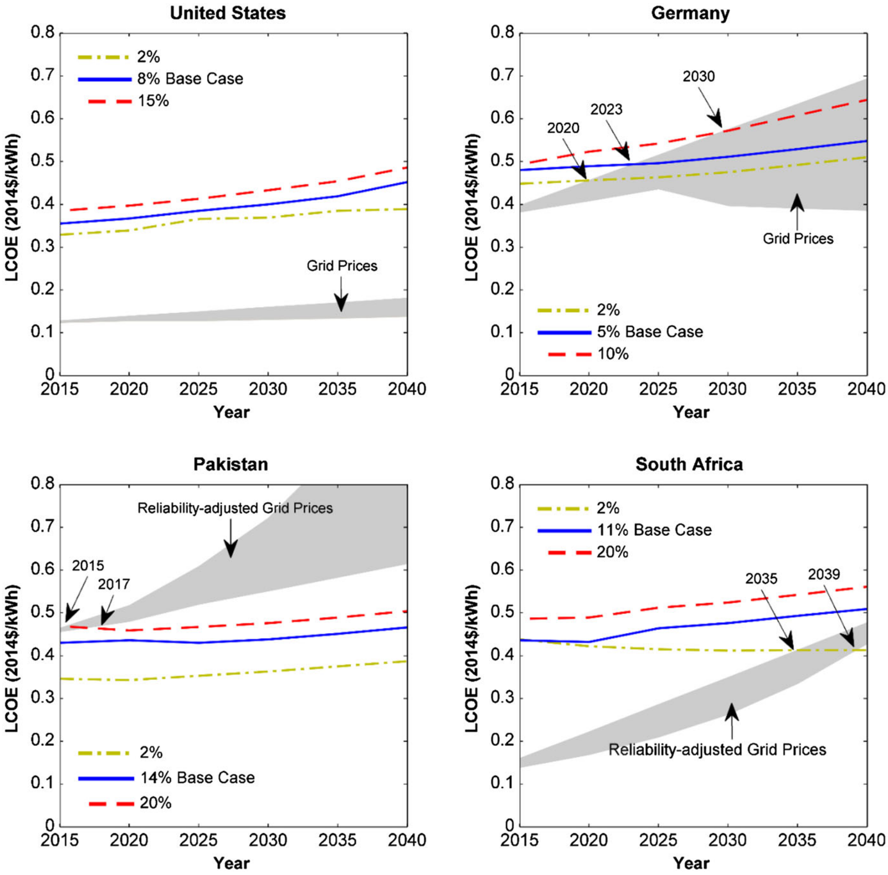

Fig. 6 Impact of changes in the cost of capital rate on the ESM LCOE and grid parity points for all four locations. Lower discount rates lower the final LCOE and push parity estimates earlier in time. For South Africa and Pakistan, given the currently high interest rates, the effect of a policy intervention addressing the cost of financing and risk is much greater than in the developed markets

cial markets [4]. With significantly different electric load/usage patterns, solar resource profiles and the costs associated with the technologies across the four countries, the LCOE helps capture this diversity. However, the microgrid LCOE results show a similar trend in time for all locations, with a rough increase of 10 cents/kWh in each location over the 25-year study period. This suggests that, given the underlying assumptions, the levelized costs of electricity for a stand-alone solar PV/diesel/battery hybrid microgrid are likely to follow similar trends in time for different energy markets. However, this expectation is strongly dependent on regional and local policies which have a pivotal role in defining the course taken by this technology.

Estimated LCOE is relatively constant in each location and has lower variability across locations than grid electricity prices. This suggests that grid parity for microgrids will be strongly dependent on the retail price or the effective cost of grid electricity at a given location. For places like Pakistan and Germany, grid power prices are high

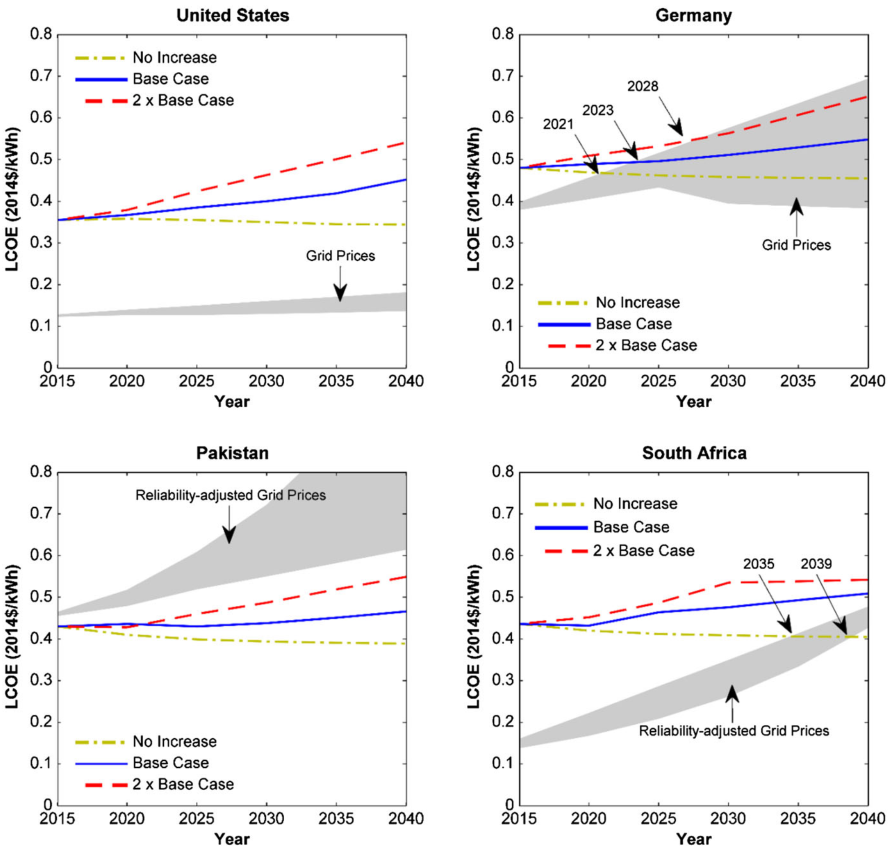

Fig. 7 Diesel price sensitivity results for all four locations. Higher fuel prices translate to higher LCOE values for all locations. 'High' diesel fuel price estimates represented by red lines are obtained by using a real increase in fuel prices twice the percent used in the base case. For the 'low' estimates, real diesel prices from the year 2015 are assumed to continue till the year 2040.A significant impact of lower diesel prices is experienced in South Africa where grid parity estimates are pushed earlier to the 2034-2039 period. System configuration details are presented in Appendix G.2 (pp. 35–38)

enough for microgrids to hit parity in the near term, as opposed to the scenario in South Africa or the United States. To illustrate the importance of grid electricity prices over microgrid design constraints, Fig. 8 shows microgrid costs for a system in Hawaii compared to the system in Ohio. Even though microgrid LCOE values for Hawaii and Ohio are relatively close, much higher grid prices in Hawaii result in an earlier grid parity date.

The preceding results and analysis are limited in scope to a comparison of LCOE obtained from costs that include subsidies and taxes, rather than the true social costs associated with microgrid systems. The retail electricity prices used are also the final prices faced by the end users rather than net social cost. This is done intentionally to estimate the economic decision faced by a consumer or microgrid developer.

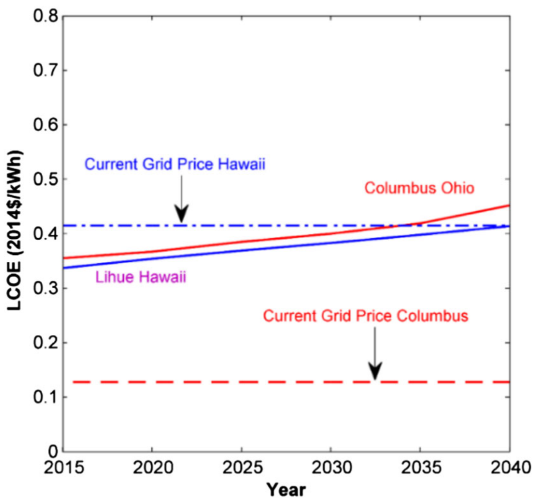

Fig. 8 ESM LCOE output for Lihue, Hawaii and Columbus, Ohio. A better solar PV resource in Hawaii does not significantly affect the final LCOE results. The *dotted line* represents the current real average electricity prices for Hawaii. Due to high grid power prices, grid parity is already here for such a stand-alone off-grid system in Hawaii

Monetization of power outages reflects the true costs of unreliable grid power in Pakistan and South Africa. The use of both the generator and batteries to determine these costs helps measure and internalize the unreliability of the grid within the cost to users. Different cost figures for the two substitutes reveal how the choice of a back-up system affects the true cost of unreliable grid power. This may form the basis for a decision like switching to microgrids. The willingness to pay for reliable power in developing countries is expected to be high, as investigated by [39]. With high effective costs of unreliable power and a greater willingness to pay, stand-alone microgrids may find a greater utility in such markets at present or in the near future.

Investigation of the optimal system configurations reveals that the cost of storage (in this case, lead acid batteries) is an important limiting factor for the fraction of solar PV used in the final system. With improving technology, this cost of storage is anticipated to fall in the coming decades. However, a comparison of optimization results with constant and significantly reduced lead acid battery prices for Germany and Pakistan in the year 2040 reveals that the cost of battery has a pronounced impact on the optimal configuration only if the solar resource is good (Table 6). Moreover, the 30% real decrease in battery prices has little impact on the LCOE of the least cost system with just 5 and 10% decrease for Germany and Pakistan respectively (Table 6).

In markets like Germany, where most of the solar PV components already have low prices, policy makers may attempt to subsidize storage as a way of increasing the proportion of solar PV in a hybrid microgrid. However, a comparison of LCOE results that include a 30% subsidy on lead acid batteries in Germany reveals that such a policy intervention may not be an effective option. The results in Table 7, show that this subsidy has little impact on the solar PV fraction (5% increase) and the LCOE (6% decrease). Considering the success of the German government's recent subsidies on battery storage for grid-tied solar PV residential systems, this is an important observation about extending that policy to off-grid systems.

Table 6 Storage cost sensitivity results for Germany and Pakistan in the year 2040

	Storage cost sensitivity	2040 Base case	2040 30% Real decrease
Germany	LCOE 2014 $/kWh	0.548	0.516
	Solar fraction %	66.4	66.4
Pakistan	LCOE 2014 $/kWh	0.466	0.4215
	Solar fraction %	60.1	90.6

A 30% real decrease in battery costs increases the solar PV fraction in Pakistan by 50%. For Germany, the fraction remains unchanged due to the limited solar resource. In either location, the impact on the LCOE is small

Table 7 A 30% subsidy on lead acid batteries in Germany has little impact on the solar PV fraction and the LCOE of the optimal system

Germany		2015
Base case	LCOE 2014 $/kWh	0.480
	Solar fraction %	53.7
30% Battery subsidy	LCOE 2014 $/kWh	0.449
	Solar fraction %	56.4

Sensitivity investigation reveals that the microgrid LCOE is fairly sensitive to diesel prices at all locations. These results suggest that places with good solar resource and cheap diesel (e.g. Saudi Arabia) make for an attractive location to set up such hybrid stand-alone microgrids, even though the resulting microgrid will get a large fraction of its energy from diesel.

Sensitivity results for the cost of capital rate reveal its importance for developing markets like South Africa and Pakistan. With currently higher lending rates, these countries have an opportunity to improve microgrid economics through financial mechanisms. This is reflected in a significant decrease in the LCOE figures with lower cost of capital rates. More importantly, this LCOE sensitivity to cost of capital rates demonstrates the heavy reliance of stand-alone microgrid technology on policies, especially in developing markets like Pakistan and South Africa. As an example, Fig. 9 compares Pakistan LCOE results for a case with lower cost of capital rates and another with lower solar PV costs due to improved system learning in Pakistan.

Increasing the learning rate from 15% to a relatively ambitious 20% reduces the LCOE values by less than 5% while a cost of capital rate of 0% pushes down the base case value to 35 cents/kWh in the year 2040. This is almost 25% lower than the base case result. This shows that even though technological progress and experience may push down system costs associated with solar PV, a decrease in the cost of capital rate has a greater impact in a market like Pakistan. With Islamic Sharia-compliant banking alternatives and their underlying principle of risk sharing [40], such low cost of capital rates could theoretically be achieved in Pakistan.

Therefore, in order to have sustainable microgrid growth, governments should expand access to finance for these systems. This result is in agreement with the findings of [23], which suggest that efforts to expand PV installation in better sunshine regions of the world may benefit greatly from policies designed to make low-cost finance more widely available. Therefore, to support microgrids, policy makers need to consider the

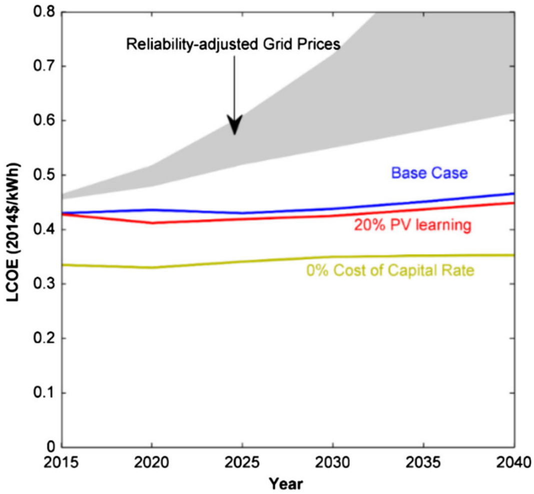

Fig. 9 The effect of a high PV learning rate vs. 0% cost of capital rate on the final LCOE for a system in Pakistan. The grid parity region depicts the effective cost of unreliable power to consumers calculated using a generator back-up system. A 20% learning rate for solar PV systems, in comparison to the 14% rate from the base case, only lowers the LCOE by an average of 5%. The cost of capital rate has a greater impact on the LCOE, indicating the effectiveness of policies focused on system financing. A 20% learning rate for solar PV is widely reported in past literature and so is used as a reference point for the comparison. System configurations are presented in Appendix G.3 (p. 39)

costs of financing as a variable that is at least as important as component-level subsidy or policy support.

In countries with relatively high (Pakistan) or steadily increasing (South Africa) effective costs of electricity to consumers, rising electricity demands and a considerable unserved rural population make stand-alone microgrids an attractive alternative. Results for Pakistan show grid parity is already here, even in areas that have electricity service. In Pakistan, more than half of the cost of reliable grid electricity is due to the cost of substitute power during the roughly 4000 h of annual power outages (approximately half the year). Factoring in the costs associated with grid extension for unserved rural areas would further raise these grid costs and only push parity to earlier dates.

For South Africa, even though the parity point is farther into the future, the possibility of it occurring earlier than the results of this analysis cannot be completely ignored. Currently, South Africa has one of the lowest electricity prices in the world due to government price controls [41]. Given the recent power shortage situation, the South African government's difficulties in implementing cost-reflective tariffs, and the decaying condition of the power sector, the country may end up in a situation similar to that of Pakistan. The government may not be able to sustain this price control for long, as it can only worsen the power outage situation. In such a pessimistic scenario, one can expect grid parity to occur sooner than suggested by these results.

The results also contribute to an evolving discussion about the true costs and benefits of off-grid systems in competition with grid electricity. Recent studies show a divided opinion over concepts like "cascading grid defection" and "utility death spiral" which state that the economics of off-grid systems would soon become favorable enough for customers to self-supply electricity and push up prices for other utility customers causing further "defection". Studies like [6] conclude that solar-plus-battery grid parity is already here or coming soon for utility customers in the US which could trigger a revenue decay for utilities, eventually leading to the demise of traditional utility business models. Others like [7] caution that grid defection is expensive relative to the low costs of grid electricity in many developed regions. Our results agree more strongly with the latter and we feel [6] use a series of assumptions (like over-optimistic interest rates, constrained use of the generator, definition of parity which skews results) that are highly favorable towards microgrid economics. Regardless, both of these recent analyses investigate microgrid cost parity in developed regions with low grid power prices (US and Australia), while we find that the situation may be significantly more favorable in other areas with higher electricity prices or lower reliability (Pakistan).

Following this investigation, it can be concluded that microgrids make more economic sense in places with expensive grid electricity and/or unreliable grids which result in a higher effective cost of grid power. While economics and grid parity are important, they may not be the only decision criteria for the adoption of such stand-alone grids. Political priorities and regulation, affinity for self-generation, reliability of such microgrid systems, the existing energy situation and consumer circumstances, the cost of capital, as well as the current share of renewables in the energy mix could govern their utility and eventual adoption in different markets. So, for rural regions without access to grid power in developing countries like Pakistan and South Africa, microgrids may be the only economic solution. For locations with long transmission distances or congested transmission infrastructure like in Germany [42], microgrids could help localize the supply of power and offer a solution. Current high power prices like those in Hawaii could drive the consumers towards self-generation using these grids. Existing levels of experience with renewables and their share in the energy mix could eventually determine the role of competing energy sources and technologies. Or a strong government stance on climate change could determine the fate of policies and these renewable-based grids in a particular market. While grid parity could mark a tipping point for these grids, their eventual adoption may be a function of many factors.

6 Conclusions and policy implications

This work attempts to determine the grid price parity dates for hybrid microgrids in four energy markets. Grid parity is important for policy makers as well as researchers, energy developers and investors working on distributed energy systems because stand-alone microgrids are expected to rely on institutional and government support. Moreover, grid price parity dates for a market may form the basis for important decisions as these emerging microgrids approach price parity.

Our results reveal that stand-alone solar PV hybrid microgrids are already cheaper than reliability-adjusted grid prices in Pakistan by about 3 cents/kWh, and they may compete with German grid prices in 10–20 years. Investigation of the US and South

African energy markets shows that, due to electricity prices as low as 10 cents/kWh, off-grid microgrids will remain more expensive than grid electricity for the foreseeable future. While microgrid LCOE estimates vary by location (e.g. 35–48 cents/kWh for the year 2015), the cost of microgrid electricity is expected to rise slowly in each location (e.g. by 10 cents/kWh over the 25 years for the US) due to increasing diesel prices and the inability of solar PV cost declines to offset fuel expenditures. Thus, microgrid price parity relies more on prevailing retail electricity prices and reliability than solar prices, solar resource, or other factors that dictate microgrid design.

With different applications in both developing and developed countries, consequences of grid parity could be different in each market. For developed countries, price parity could encourage consumers to self-supply electricity and "defect" from the main grid, pushing up prices for other utility customers causing further "defection" [43]. In developing countries, once microgrids hit parity, they could help improve access to power by providing energy to remote areas or offer an alternative to traditional policy measures of load shedding or construction of large-scale central generation. In either case, grid parity could result in a paradigm-shift which may have important consequences for the energy sectors around the world.

Analysis of our results reveals that grid parity for microgrids is a function of policies and not just technological advancements. Therefore, in heavily regulated markets like Pakistan such stand-alone systems may only make headway following important policy reforms. Our sensitivity analysis results demonstrate the impact and importance of the cost of financing to improve microgrid competitiveness, particularly in developing countries like Pakistan and South Africa where dropping to a 2% cost of capital rate reduces the 2035 base case LCOE values by 8 cents/kWh (approximately 20%). Therefore, policies aimed at attracting lower-interest investment can play a pivotal role in the adoption of microgrids.

Following from these observations, it can be concluded that such stand-alone microgrids may find applications sooner in developing countries with problems like energy shortage and limited access to grid power. However, most of these countries follow energy policy regimes adopted in the developed world. This leaves them at a disadvantage if the industrialized world does not establish model microgrid policies first. With a lack of necessity for formulating sophisticated microgrid policies in the developed world, developing countries need to take on the role of policy innovators to effectively make use of these microgrids to solve some of their woes.

With an aging infrastructure failing to meet demand in places like Pakistan and South Africa, there already appears to be a growing need for electricity policy reforms. With relatively high effective costs of unreliable electricity and lots of sunny days in the year, they have a greater opportunity to make the most of renewable based hybrid microgrids, but cannot do this without a shift in policy. These countries, and others like them, need clear policies and sound implementation mechanisms that create a supportive environment for microgrids. Focusing on financing for microgrid projects is important to attract investment and reduce economic risks. Providing capital subsidies to cover the initial costs or giving legal cover to third party service providers through efficient licensing frameworks can encourage investment. The 2003 Electricity Act in India is a good example of a government initiative which deregulated tariffs and allowed third party service providers to set up microgrids in specific areas [44].

Enabling private and third party financing through government guarantees, ensuring low-interest loans, standardized power sales contracts and production incentives can all help bring down the cost of capital. At the same time, educating stakeholders and institutionalizing to ensure the implementation mechanism is robust is also important. The standard rural electrification scheme (F.A.C.E) encouraged by the French government for hybrid PV systems is one such example [45]. Additionally, commissioning studies and research to determine the best business models for microgrids can be useful. These steps can create a conducive environment for microgrids in places where the technology is close to parity or is already economically viable.

This work concludes that stand-alone renewable based microgrids make more economic sense in places with unreliable grids and/or high grid power prices. However, existing policy frameworks play a key role in determining their ultimate fate. This is amply demonstrated by our parity results for Pakistan where, despite their immediate economic value, existing policies and regulations have blocked out these stand-alone microgrids. Therefore, it is important for developing countries to consider these systems as valuable policy alternatives to the conventional approaches. For the developed world, unless grid prices are high, "grid defection" in such markets is primarily going to be a choice rather than a necessity.

Future work could expand upon our efforts in several ways. Because of our focus on detailed examination of each technology, we limited our work to solar PV/diesel/lead-acid microgrids. Using the same methods, expansion to other generation or storage technologies (such as wind or lithium-ion batteries) would help survey the costs of alternative designs. Microgrid modeling can also be improved, through more sophisticated battery operation algorithms, and better integration of uncertainty into the system design and operation models. Finally, the effects on electricity systems from adoption of distributed energy resources and microgrids is a critical question that is only now being investigated in literature. While our work assumes an extrapolation of current trends for electricity grids, it is clear that broad adoption of distributed energy technologies will change the electricity industry, though the exact nature of those changes is yet unknown.

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